Sparse Parameterization for Epitomic Dataset Distillation

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Dataset distillation

Dataset distillation aims to obtain smaller synthetic datasets that capture the essential information of the originals.

$$\mathcal{S}^* = \arg\min_{\mathcal{S}} \mathbb{E}_{\theta\sim\Theta}\left[\mathcal{D}\left(\varphi(\mathcal{T},\theta),\varphi(\mathcal{S},\theta)\right)\right]$$

The success of dataset distillation relies on two essential factors:

- Proper parameterization of the synthetic dataset
- Effective design of the matching objective to align it with the original

Previous explorations

Spatial redundancy exists within individual synthetic images and between different synthetic images.

- Naïve image-independent parameterization: each learnable parameter basis (synthetic image) is optimized independently
- Recent efficient parameterization: either only consider the connections between synthetic images, or do not reduce the redundancy inside the synthetic image thoroughly

Sparse parameterization

Our sparse parameterization strategy leverages principles from representation learning paradigms, including convolutional neural network, vision transformer, dictionary learning, and sparse coding. The data synthesis can be formulated as:

$$\tilde{X}_i = \Phi_\phi(E, A_i)$$

- Spatial-agnostic epitomic tokens (SAETs) E are shared among all patches in all synthetic images, reducing spatial redundancy
- Sparse coding matrices (SCMs) A_i are stored in a sparse manner, eliminating inefficient parameters and bringing better generalization
- Feature-recurrent network (FReeNet) Φ_ϕ recurrently use the sparsely combined features, alleviating storage burden

Multi-head sparse coding:

$$\mathrm{MHSC}_r\left(\{E^h\}_{h=1}^H, \{A_i^h\}_{h=1}^H\right) = W_r[E^1A_i^1, E^2A_i^2, ..., E^HA_i^H] + b_r$$

Non-linear transformation:

$$Z_r' = \mathrm{LN}_r^1(\mathrm{MHSC}_r\left(\{E^h\}_{h=1}^H, \{A_i^h\}_{h=1}^H\right)) + Z_{r-1}), \quad r = 1, 2, ..., R,$$

$$Z_r = \mathrm{LN}_r^2(\mathrm{MLP}_r(Z_r') + Z_r'), \quad r = 1, 2, ..., R,$$

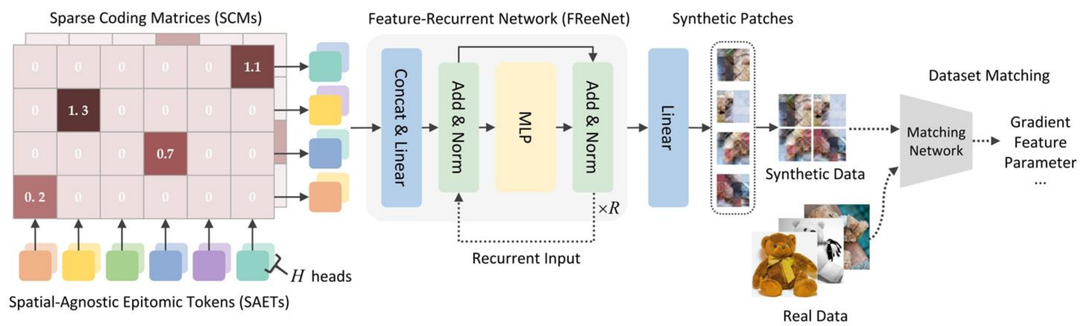
$$\tilde{X}_i = WZ_R + b,$$

Training objective with sparsity penalty:

$$\mathcal{S} = \left\{(\Phi_\phi(\{E^h\}_{h=1}^H, \{A_i^h\}_{h=1}^H), y_i)\right\}_{i=1}^N,$$

$$\arg\min_{\{E^h\}_{h=1}^H, \{\{A_i^h\}_{h=1}^H\}_{i=1}^N, \phi} \mathbb{E}_{\theta\sim\Theta}\left[\mathcal{D}\left(\varphi(\mathcal{T},\theta),\varphi(\mathcal{S},\theta)\right)\right] + \lambda\sum_{i=1}^N\sum_{h=1}^H ||A_i^h||_1,$$

Overview



Storage analysis

- Coordinate sparse matrix storage format (COO) for SCMs
- Storage equation under IPC=n storage budget

$$\underbrace{DK}_{\text{SAETs: }\{E^h\}_{h=1}^H} + \underbrace{1.5NHk}_{\text{SCMs: }\{\{\bar{A}_i^h\}_{h=1}^H\}_{i=1}^N} + \underbrace{R(3D^2+7D)+L(D+1)}_{\text{FReeNet: }\phi} \le \underbrace{cnLJ}_{\text{Budget: IPC}=n}.$$

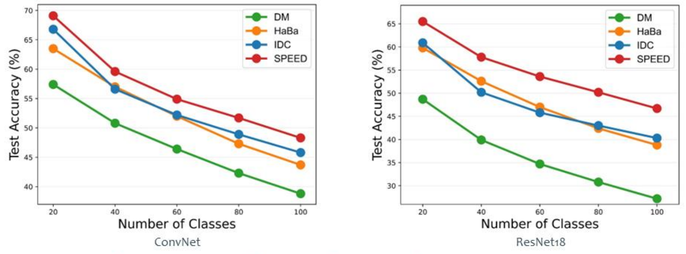
Comparisons on standard benchmarks

	Dataset	CIFAR10			CIFAR100			TinyImageNet		
	IPC	1	10	50	1	10	50	1	10	50
Coreset	Random	14.4±2.0	26.0±1.2	43.4±1.0	4.2±0.3	14.6±0.5	30.0±0.4	1.4±0.1	5.0±0.2	15.0±0.4
	Herding	21.5±1.3	31.6±0.7	40.4±0.6	8.4±0.3	17.3±0.3	**33.7±0.5**	**2.8±0.2**	6.3±0.2	**16.7±0.3**
	K-Center	**23.3±0.9**	**36.4±0.6**	**48.7±0.3**	**8.6±0.3**	**20.7±0.2**	33.6±0.4	2.7±0.2	**7.8±0.4**	16.7±0.4
	Forgetting	13.5±1.2	23.3±1.0	23.3±1.1	4.5±0.3	9.8±0.2	-	1.6±0.1	5.1±0.2	15.0±0.3
Matching	DC [13]	28.3±0.5	44.9±0.5	53.9±0.5	12.8±0.3	25.2±0.3	32.1±0.3	-	-	-
	DSA [14]	28.8±0.7	52.1±0.5	60.6±0.5	13.9±0.3	32.3±0.3	42.8±0.4	-	-	-
	KIP [47]	**49.9±0.2**	62.7±0.3	68.6±0.2	15.7±0.2	28.3±0.1	-	-	-	-
	DM [17]	26.0±0.8	48.9±0.6	63.0±0.4	11.4±0.3	29.7±0.3	43.6±0.4	3.9±0.2	12.9±0.4	24.1±0.3
	TM [16]	46.3±0.8	65.3±0.7	71.6±0.2	24.3±0.3	40.1±0.4	**47.7±0.2**	8.8±0.3	23.2±0.2	**28.0±0.3**
	FRePo [37]	46.8±0.7	**65.5±0.4**	**71.7±0.2**	**28.7±0.1**	**42.5±0.2**	44.3±0.2	**15.4±0.3**	**25.4±0.2**	-
Parameters / Class		3,072	30,720	153,600	3,072	30,720	153,600	12,288	122,880	614,400
Param.	IDC [19]	50.0±0.4	67.5±0.5	74.5±0.1	-	44.8±0.2	-	-	-	-
	HaBa [21]	48.3±0.8	69.9±0.4	74.0±0.2	33.4±0.4	40.2±0.2	47.0±0.2	-	-	-
	RTP [22]	**66.4±0.4**	71.2±0.4	73.6±0.5	34.0±0.4	42.9±0.7	-	16.0±0.7	-	-
	SPEED (Ours)	63.2±0.1	**73.5±0.2**	**77.7±0.4**	**40.0±0.4**	**45.9±0.3**	**49.1±0.2**	**26.9±0.3**	**28.8±0.2**	**30.1±0.3**
Whole Dataset		84.8±0.1			56.2±0.3			37.6±0.4		

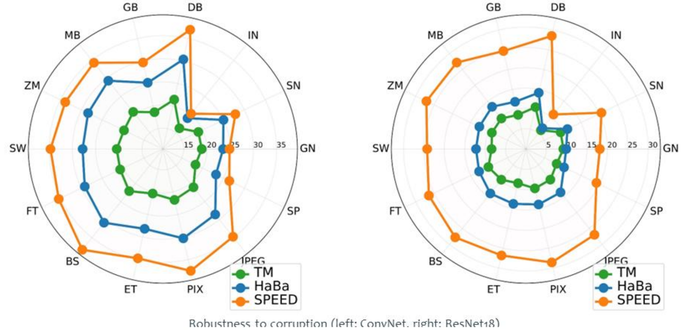
Comparisons with previous dataset distillation and coreset selection methods on standard benchmarks

Synthetic images

Application in continual learning



Generalization of synthetic dataset



Robustness to corruption (left: ConvNet, right: ResNet18)

Method	ConvNet	MLP	ResNet18	ViT
IDC [19]	67.5±0.5	41.4±0.2	62.9±0.6	47.9±0.8
HaBa [21]	69.9±0.4	35.4±0.4	60.2±0.9	42.2±0.6
RTP [22]	71.2±0.4	27.2±0.2	67.5±0.1	35.7±0.4
SPEED	**73.5±0.2**	**44.4±0.4**	**69.5±0.4**	**51.5±0.3**

Cross-architecture performance